Superstore EDA — Final Portfolio Notebook

Case: Superstore Retail Transactions
Notebook purpose: audit-ready exploratory analysis for a retail sales dataset
Lifecycle alignment: ASK → PREPARE → PROCESS → ANALYZE

Notebook rules

- Validate before analyzing
- Do not overclaim causality
- Keep all transformations explicit
- Record assumptions, limitations, and validation checks

1. Analytical objective

This notebook is designed to answer the following non-causal business questions:

1. Which categories, sub-categories, regions, and customer segments contribute most to sales and profit?
2. Where are the main negative-profit patterns concentrated?
3. How are discounting, sales, and profitability associated at a descriptive level?

Scope

- Descriptive and exploratory analysis only
- No forecasting
- No causal inference
- No recommendation inflation beyond the evidence

Assumptions

- Each row represents one order line
- `Row ID` is intended to be unique
- `Sales`, `Quantity`, `Discount`, and `Profit` are transaction-level measures
- `Order Date` and `Ship Date` are parseable dates

Limitations

- This is a static dataset, not a live retail system
- Results depend on dataset quality and may not generalize to other retailers
- Associations in this notebook do not establish business causality

2. Environment setup

In [95]:
```
# --- Environment Setup ---
import warnings
warnings.filterwarnings("ignore")

from pathlib import Path
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# --- Display Settings ---
pd.set_option("display.max_columns", None)
pd.set_option("display.width", 140)
pd.set_option("display.float_format", lambda x: f"{x:,.2f}")

# --- Reproducibility ---
RANDOM_STATE = 42

# --- Project Paths ---
PROJECT_ROOT = Path.cwd().parent
DATA_PATH = PROJECT_ROOT / "data" / "raw" / "Sample - Superstore.csv"

print(f"Project root: {PROJECT_ROOT}")
print(f"Expected data path: {DATA_PATH}")
```

```
Project root: c:\Users\Lazarus\Github\data-analysis-lab
Expected data path: c:\Users\Lazarus\Github\data-analysis-lab\data\raw\Sample - Supe
rstore.csv
```

3. Data load

In [96]:
```
# --- Data Load (Validated) ---
if not DATA_PATH.exists():
    raise FileNotFoundError(
        f"\nDataset not found at:\n{DATA_PATH}\n\n"
        "Fix options:\n"
        "1. Move file to data/raw/\n"
        "2. Or update DATA_PATH\n"
    )

df = pd.read_csv(DATA_PATH, encoding="latin1")

print(f"Loaded dataset successfully")
print(f"Shape: {df.shape[0]:,} rows × {df.shape[1]} columns")
```

```
Loaded dataset successfully
Shape: 9,994 rows × 21 columns
```

4. PREPARE — structural validation

This section documents the raw dataset structure without changing the data.

In [97]:
```
prepare_summary = {
    "row_count": len(df),
    "column_count": df.shape[1],
    "columns": df.columns.tolist()
}
prepare_summary
```

Out[97]:
```
{'row_count': 9994,
 'column_count': 21,
 'columns': ['Row ID',
  'Order ID',
  'Order Date',
  'Ship Date',
  'Ship Mode',
  'Customer ID',
  'Customer Name',
  'Segment',
  'Country',
  'City',
  'State',
  'Postal Code',
  'Region',
  'Product ID',
  'Category',
  'Sub-Category',
  'Product Name',
  'Sales',
  'Quantity',
  'Discount',
  'Profit']}
```

In [98]:
```
schema_df = pd.DataFrame({
    "column": df.columns,
    "observed_dtype": df.dtypes.astype(str).values,
    "non_null_count": df.notna().sum().values,
    "null_count": df.isna().sum().values,
    "null_pct": ((df.isna().sum() / len(df)) * 100).round(2).values,
    "n_unique": df.nunique(dropna=False).values
}).sort_values(["null_pct", "column"], ascending=[False, True])

schema_df
```

Out[98]:

	column	observed_dtype	non_null_count	null_count	null_pct	n_unique
14	Category	str	9994	0	0.00	3
9	City	str	9994	0	0.00	531
8	Country	str	9994	0	0.00	1
5	Customer ID	str	9994	0	0.00	793
6	Customer Name	str	9994	0	0.00	793
19	Discount	float64	9994	0	0.00	12
2	Order Date	str	9994	0	0.00	1237
1	Order ID	str	9994	0	0.00	5009
11	Postal Code	int64	9994	0	0.00	631
13	Product ID	str	9994	0	0.00	1862
16	Product Name	str	9994	0	0.00	1850
20	Profit	float64	9994	0	0.00	7287
18	Quantity	int64	9994	0	0.00	14
12	Region	str	9994	0	0.00	4
0	Row ID	int64	9994	0	0.00	9994
17	Sales	float64	9994	0	0.00	5825
7	Segment	str	9994	0	0.00	3
3	Ship Date	str	9994	0	0.00	1334
4	Ship Mode	str	9994	0	0.00	4
10	State	str	9994	0	0.00	49
15	Sub-Category	str	9994	0	0.00	17

In [99]:

```
print("Sample rows:")
display(df.sample(5, random_state=RANDOM_STATE))
```

```
Sample rows:
```

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Co
3125	3126	CA-2015-121720	6/11/2015	6/12/2015	First Class	JE-15610	Jim Epp	Corporate	U
1441	1442	CA-2017-128160	12/19/2017	12/24/2017	Second Class	MM-17920	Michael Moore	Consumer	U
4510	4511	CA-2016-119935	11/10/2016	11/14/2016	Standard Class	KM-16225	Kalyca Meade	Corporate	U
39	40	CA-2015-117415	12/27/2015	12/31/2015	Standard Class	SN-20710	Steve Nguyen	Home Office	U
4509	4510	CA-2017-167003	5/23/2017	5/29/2017	Standard Class	VS-21820	Vivek Sundaresam	Consumer	U

4.1 Grain and key checks

In [100…

```
candidate_key_checks = {
    "row_id_unique": df["Row ID"].is_unique,
    "duplicate_full_rows": int(df.duplicated().sum()),
    "duplicate_row_id": int(df["Row ID"].duplicated().sum()),
    "duplicate_order_product_pairs": int(df.duplicated(subset=["Order ID", "Product
}
candidate_key_checks
```

Out[100…

```
{'row_id_unique': True,
 'duplicate_full_rows': 0,
 'duplicate_row_id': 0,
 'duplicate_order_product_pairs': 8}
```

4.2 Missingness profile

In [69]:
```
missing_df = (
    pd.DataFrame({
        "missing_count": df.isna().sum(),
        "missing_pct": (df.isna().sum() / len(df) * 100).round(2)
    })
    .sort_values(["missing_pct", "missing_count"], ascending=False)
)

missing_df[missing_df["missing_count"] > 0]
```

Out[69]:

	missing_count	missing_pct

4.3 Type and range validation

In [70]:
```
numeric_cols = ["Sales", "Quantity", "Discount", "Profit"]
range_checks = {
    "negative_sales_count": int((df["Sales"] < 0).sum()),
    "non_positive_quantity_count": int((df["Quantity"] <= 0).sum()),
    "discount_below_zero_count": int((df["Discount"] < 0).sum()),
    "discount_above_one_count": int((df["Discount"] > 1).sum()),
    "negative_profit_count": int((df["Profit"] < 0).sum()),
}
range_checks
```

Out[70]:
```
{'negative_sales_count': 0,
 'non_positive_quantity_count': 0,
 'discount_below_zero_count': 0,
 'discount_above_one_count': 0,
 'negative_profit_count': 1871}
```

In [71]:
```
df["Order Date Parsed"] = pd.to_datetime(df["Order Date"], errors="coerce")
df["Ship Date Parsed"] = pd.to_datetime(df["Ship Date"], errors="coerce")

date_checks = {
    "invalid_order_date_count": int(df["Order Date Parsed"].isna().sum()),
    "invalid_ship_date_count": int(df["Ship Date Parsed"].isna().sum()),
    "ship_before_order_count": int((df["Ship Date Parsed"] < df["Order Date Parsed"
    "min_order_date": df["Order Date Parsed"].min(),
    "max_order_date": df["Order Date Parsed"].max(),
}
date_checks
```

Out[71]:
```
{'invalid_order_date_count': 0,
 'invalid_ship_date_count': 0,
 'ship_before_order_count': 0,
 'min_order_date': Timestamp('2014-01-03 00:00:00'),
 'max_order_date': Timestamp('2017-12-30 00:00:00')}
```

In [72]:
```
categorical_scan = {
    "segments": sorted(df["Segment"].dropna().unique().tolist()),
    "regions": sorted(df["Region"].dropna().unique().tolist()),
    "categories": sorted(df["Category"].dropna().unique().tolist()),
    "ship_modes": sorted(df["Ship Mode"].dropna().unique().tolist())
```

```
}
categorical_scan
```

Out[72]:
```
{'segments': ['Consumer', 'Corporate', 'Home Office'],
 'regions': ['Central', 'East', 'South', 'West'],
 'categories': ['Furniture', 'Office Supplies', 'Technology'],
 'ship_modes': ['First Class', 'Same Day', 'Second Class', 'Standard Class']}
```

4.4 PREPARE validation summary

In [73]:
```
prepare_validation = pd.DataFrame([
    ["Row count confirmed", True],
    ["Column count confirmed", True],
    ["Schema documented", True],
    ["Missingness profiled", True],
    ["Duplicate checks performed", True],
    ["Range checks performed", True],
    ["Date parsing checks performed", True],
    ["No transformations applied to core business fields yet", True],
], columns=["check", "status"])

prepare_validation
```

Out[73]:

	check	status
0	Row count confirmed	True
1	Column count confirmed	True
2	Schema documented	True
3	Missingness profiled	True
4	Duplicate checks performed	True
5	Range checks performed	True
6	Date parsing checks performed	True
7	No transformations applied to core business fi...	True

5. PROCESS — explicit cleaning and feature engineering

Only minimal, transparent changes are applied here.
Raw business columns are preserved where practical.

In [74]:
```
df_clean = df.copy()

# Preserve parsed dates as analysis-ready fields
df_clean = df_clean.rename(columns={
    "Order Date Parsed": "order_date",
    "Ship Date Parsed": "ship_date"
```

```
})

# Standardized helper columns
df_clean["year"] = df_clean["order_date"].dt.year
df_clean["month"] = df_clean["order_date"].dt.to_period("M").astype(str)
df_clean["ship_delay_days"] = (df_clean["ship_date"] - df_clean["order_date"]).dt.d

processing_log = pd.DataFrame([
    ["Create parsed order date", "Order Date -> order_date", "Supports temporal val
    ["Create parsed ship date", "Ship Date -> ship_date", "Supports shipping lag va
    ["Create year", "order_date.dt.year", "Supports yearly aggregation"],
    ["Create month", "order_date.dt.to_period('M')", "Supports monthly aggregation"
    ["Create ship delay", "ship_date - order_date", "Supports fulfillment descripti
], columns=["step", "change", "rationale"])

processing_log
```

Out[74]:

	step	change	rationale
0	Create parsed order date	Order Date -> order_date	Supports temporal validation and analysis
1	Create parsed ship date	Ship Date -> ship_date	Supports shipping lag validation and analysis
2	Create year	order_date.dt.year	Supports yearly aggregation
3	Create month	order_date.dt.to_period('M')	Supports monthly aggregation
4	Create ship delay	ship_date - order_date	Supports fulfillment descriptives

In [75]:

```
process_validation = {
    "row_count_before": len(df),
    "row_count_after": len(df_clean),
    "column_count_before": df.shape[1],
    "column_count_after": df_clean.shape[1],
    "rows_removed": len(df) - len(df_clean),
    "remaining_invalid_order_dates": int(df_clean["order_date"].isna().sum()),
    "remaining_invalid_ship_dates": int(df_clean["ship_date"].isna().sum()),
}
process_validation
```

Out[75]:

```
{'row_count_before': 9994,
 'row_count_after': 9994,
 'column_count_before': 23,
 'column_count_after': 26,
 'rows_removed': 0,
 'remaining_invalid_order_dates': 0,
 'remaining_invalid_ship_dates': 0}
```

6. ANALYZE — overview metrics

In [76]:

```
overview = pd.DataFrame({
    "metric": [
        "Orders (unique)",
```

```
        "Customers (unique)",
        "Products (unique)",
        "Total Sales",
        "Total Profit",
        "Average Sales per Row",
        "Average Profit per Row",
        "Negative Profit Rows",
        "Average Discount",
        "Median Ship Delay (days)"
    ],
    "value": [
        df_clean["Order ID"].nunique(),
        df_clean["Customer ID"].nunique(),
        df_clean["Product ID"].nunique(),
        df_clean["Sales"].sum(),
        df_clean["Profit"].sum(),
        df_clean["Sales"].mean(),
        df_clean["Profit"].mean(),
        (df_clean["Profit"] < 0).sum(),
        df_clean["Discount"].mean(),
        df_clean["ship_delay_days"].median()
    ]
})

overview
```

Out[76]:

	metric	value
0	Orders (unique)	5,009.00
1	Customers (unique)	793.00
2	Products (unique)	1,862.00
3	Total Sales	2,297,200.86
4	Total Profit	286,397.02
5	Average Sales per Row	229.86
6	Average Profit per Row	28.66
7	Negative Profit Rows	1,871.00
8	Average Discount	0.16
9	Median Ship Delay (days)	4.00

7. Sales and profit by category

In [77]:

```
category_perf = (
    df_clean.groupby("Category", as_index=False)
    .agg(
        sales=("Sales", "sum"),
        profit=("Profit", "sum"),
        orders=("Order ID", "nunique"),
```

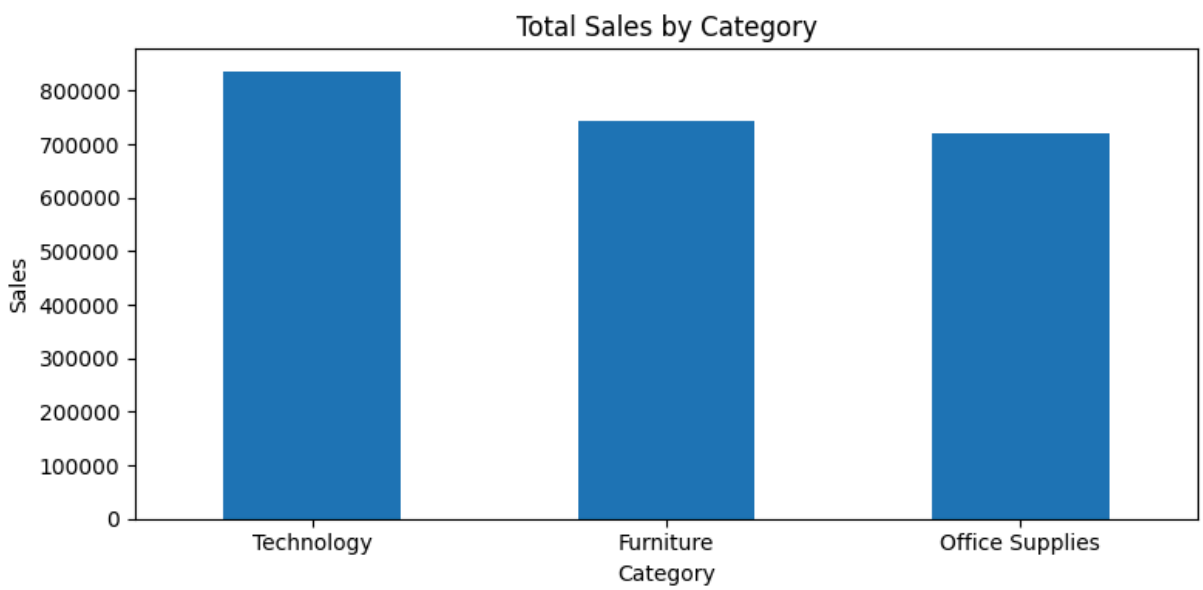
```
        avg_discount=("Discount", "mean")
    )
    .sort_values("sales", ascending=False)
)

category_perf
```

Out[77]:

	Category	sales	profit	orders	avg_discount
2	Technology	836,154.03	145,454.95	1544	0.13
0	Furniture	741,999.80	18,451.27	1764	0.17
1	Office Supplies	719,047.03	122,490.80	3742	0.16

In [78]:

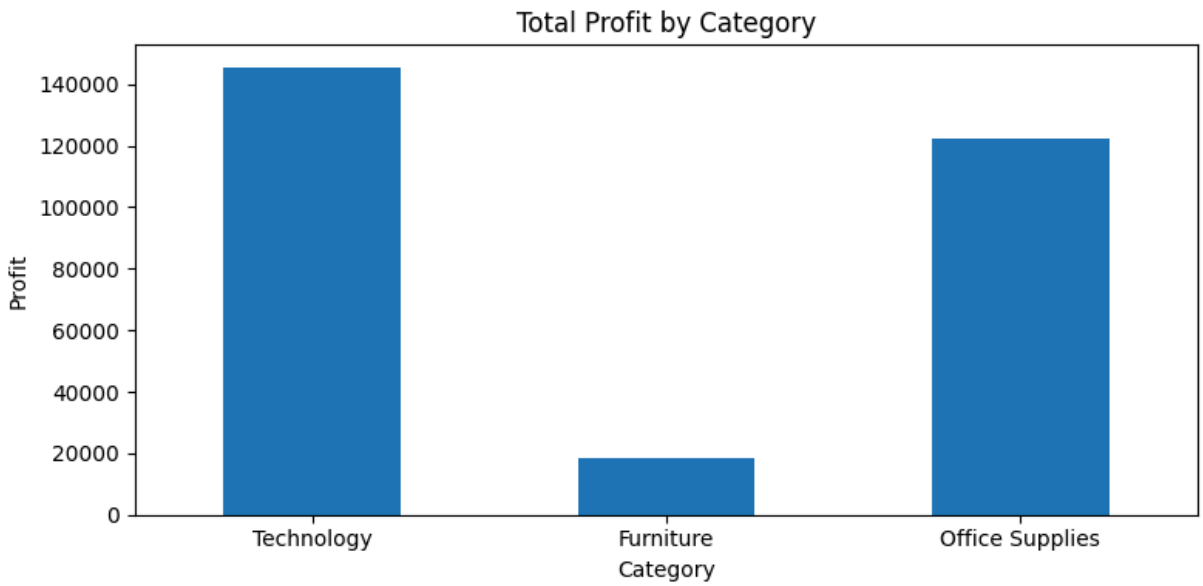
```
ax = category_perf.plot(
    x="Category",
    y="sales",
    kind="bar",
    legend=False,
    figsize=(8, 4),
    title="Total Sales by Category",
    rot=0
)
ax.set_ylabel("Sales")
plt.tight_layout()
plt.show()
```



In [79]:

```
ax = category_perf.plot(
    x="Category",
    y="profit",
    kind="bar",
    legend=False,
    figsize=(8, 4),
    title="Total Profit by Category",
    rot=0
)
```

```
ax.set_ylabel("Profit")
plt.tight_layout()
plt.show()
```



8. Top Contributors (Pareto Analysis)

In [80]:
```
subcat_sales_pareto = (
    df_clean.groupby("Sub-Category", as_index=False)
    .agg(sales=("Sales", "sum"))
    .sort_values("sales", ascending=False)
)

subcat_sales_pareto["sales_share_pct"] = (
    subcat_sales_pareto["sales"] / subcat_sales_pareto["sales"].sum() * 100
)

subcat_sales_pareto["cum_sales_share_pct"] = subcat_sales_pareto["sales_share_pct"]
subcat_sales_pareto["rank"] = range(1, len(subcat_sales_pareto) + 1)

top_20pct_n = max(1, int(np.ceil(len(subcat_sales_pareto) * 0.20)))
top_20pct_subcats = subcat_sales_pareto.head(top_20pct_n)

print("Top 20% of sub-categories by sales contribution")
display(top_20pct_subcats)

print(f"Cumulative sales share captured: {top_20pct_subcats['sales_share_pct'].sum(
```

```
Top 20% of sub-categories by sales contribution
```

	Sub-Category	sales	sales_share_pct	cum_sales_share_pct	rank
13	Phones	330,007.05	14.37	14.37	1
5	Chairs	328,449.10	14.30	28.66	2
14	Storage	223,843.61	9.74	38.41	3
16	Tables	206,965.53	9.01	47.42	4

```
Cumulative sales share captured: 47.42%
```

Interpretation:

This table identifies the top 20% of sub-categories ranked by revenue contribution and shows how much of total sales they capture.

9. Sub-category concentration and loss-making patterns

In [81]:
```
subcat_perf = (
    df_clean.groupby(["Category", "Sub-Category"], as_index=False)
    .agg(
        sales=("Sales", "sum"),
        profit=("Profit", "sum"),
        quantity=("Quantity", "sum"),
        avg_discount=("Discount", "mean")
    )
)

top_subcats_by_sales = subcat_perf.sort_values("sales", ascending=False).head(10)
bottom_subcats_by_profit = subcat_perf.sort_values("profit", ascending=True).head(1

print("Top 10 sub-categories by sales")
display(top_subcats_by_sales)

print("Bottom 10 sub-categories by profit")
display(bottom_subcats_by_profit)
```

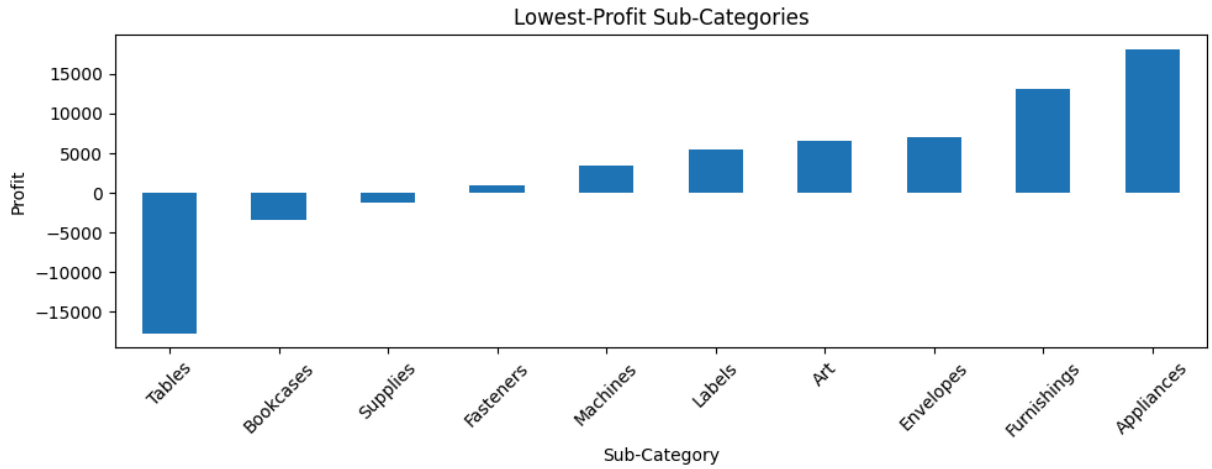
```
Top 10 sub-categories by sales
```

	Category	Sub-Category	sales	profit	quantity	avg_discount
16	Technology	Phones	330,007.05	44,515.73	3289	0.15
1	Furniture	Chairs	328,449.10	26,590.17	2356	0.17
11	Office Supplies	Storage	223,843.61	21,278.83	3158	0.07
3	Furniture	Tables	206,965.53	-17,725.48	1241	0.26
6	Office Supplies	Binders	203,412.73	30,221.76	5974	0.37
15	Technology	Machines	189,238.63	3,384.76	440	0.31
13	Technology	Accessories	167,380.32	41,936.64	2976	0.08
14	Technology	Copiers	149,528.03	55,617.82	234	0.16
0	Furniture	Bookcases	114,880.00	-3,472.56	868	0.21
4	Office Supplies	Appliances	107,532.16	18,138.01	1729	0.17

```
Bottom 10 sub-categories by profit
```

	Category	Sub-Category	sales	profit	quantity	avg_discount
3	Furniture	Tables	206,965.53	-17,725.48	1241	0.26
0	Furniture	Bookcases	114,880.00	-3,472.56	868	0.21
12	Office Supplies	Supplies	46,673.54	-1,189.10	647	0.08
8	Office Supplies	Fasteners	3,024.28	949.52	914	0.08
15	Technology	Machines	189,238.63	3,384.76	440	0.31
9	Office Supplies	Labels	12,486.31	5,546.25	1400	0.07
5	Office Supplies	Art	27,118.79	6,527.79	3000	0.07
7	Office Supplies	Envelopes	16,476.40	6,964.18	906	0.08
2	Furniture	Furnishings	91,705.16	13,059.14	3563	0.14
4	Office Supplies	Appliances	107,532.16	18,138.01	1729	0.17

In [82]:
```
ax = bottom_subcats_by_profit.sort_values("profit").plot(
    x="Sub-Category",
    y="profit",
    kind="bar",
    legend=False,
    figsize=(10, 4),
    title="Lowest-Profit Sub-Categories",
    rot=45
)
ax.set_ylabel("Profit")
plt.tight_layout()
plt.show()
```



10. Concentration — Segment Level

In [83]:
```
segment_subcat_profit = (
    df_clean.groupby(["Segment", "Sub-Category"], as_index=False)
    .agg(
        sales=("Sales", "sum"),
        profit=("Profit", "sum"),
        orders=("Order ID", "nunique"),
        avg_discount=("Discount", "mean")
    )
    .sort_values("profit", ascending=True)
)

print("Lowest-profit segment × sub-category combinations")
display(segment_subcat_profit.head(15))
```

```
Lowest-profit segment × sub-category combinations
```

	Segment	Sub-Category	sales	profit	orders	avg_discount
16	Consumer	Tables	99,933.79	-9,728.04	155	0.27
33	Corporate	Tables	70,871.72	-4,906.50	101	0.26
4	Consumer	Bookcases	68,632.73	-4,435.64	128	0.22
50	Home Office	Tables	36,160.02	-3,090.94	51	0.25
15	Consumer	Supplies	25,741.50	-1,657.55	98	0.09
42	Home Office	Fasteners	560.05	120.81	44	0.07
49	Home Office	Supplies	1,496.76	129.53	29	0.06
25	Corporate	Fasteners	783.29	251.90	58	0.12
38	Home Office	Bookcases	12,241.34	324.63	32	0.16
32	Corporate	Supplies	19,435.28	338.93	60	0.06
45	Home Office	Machines	49,419.05	540.68	22	0.27
8	Consumer	Fasteners	1,680.94	576.80	113	0.07
21	Corporate	Bookcases	34,005.92	638.45	64	0.22
28	Corporate	Machines	60,276.76	703.02	34	0.35
44	Home Office	Labels	1,675.40	709.44	66	0.08

11. Loss Concentration — Regional Level

In [84]:
```
region_subcat_profit = (
    df_clean.groupby(["Region", "Sub-Category"], as_index=False)
    .agg(
        sales=("Sales", "sum"),
        profit=("Profit", "sum"),
        orders=("Order ID", "nunique"),
        avg_discount=("Discount", "mean")
    )
    .sort_values("profit", ascending=True)
)

print("Lowest-profit region × sub-category combinations")
display(region_subcat_profit.head(15))
```

```
Lowest-profit region × sub-category combinations
```

	Region	Sub-Category	sales	profit	orders	avg_discount
33	East	Tables	39,139.81	-11,025.38	76	0.37
50	South	Tables	43,916.19	-4,623.06	49	0.22
9	Central	Furnishings	15,254.37	-3,906.22	191	0.40
16	Central	Tables	39,154.97	-3,559.65	68	0.26
1	Central	Appliances	23,582.03	-2,638.62	119	0.45
4	Central	Bookcases	24,157.18	-1,997.90	49	0.23
55	West	Bookcases	36,004.12	-1,646.51	79	0.23
11	Central	Machines	26,797.38	-1,486.07	20	0.33
45	South	Machines	53,890.96	-1,438.89	17	0.33
21	East	Bookcases	43,819.33	-1,167.63	68	0.22
32	East	Supplies	10,760.12	-1,155.14	55	0.07
3	Central	Binders	56,923.28	-1,043.64	318	0.51
15	Central	Supplies	9,467.37	-661.89	35	0.14
62	West	Machines	42,444.12	-618.93	39	0.30
49	South	Supplies	8,318.93	1.88	29	0.10

Purpose:

Localize loss-making patterns geographically.

12. Regional and segment performance

In [85]:
```
region_perf = (
    df_clean.groupby("Region", as_index=False)
    .agg(
        sales=("Sales", "sum"),
        profit=("Profit", "sum"),
        orders=("Order ID", "nunique"),
        avg_discount=("Discount", "mean")
    )
    .sort_values("sales", ascending=False)
)

segment_perf = (
    df_clean.groupby("Segment", as_index=False)
    .agg(
        sales=("Sales", "sum"),
        profit=("Profit", "sum"),
        orders=("Order ID", "nunique"),
```

```
        avg_discount=("Discount", "mean")
    )
    .sort_values("sales", ascending=False)
)

print("Region performance")
display(region_perf)

print("Segment performance")
display(segment_perf)
```

```
Region performance
```

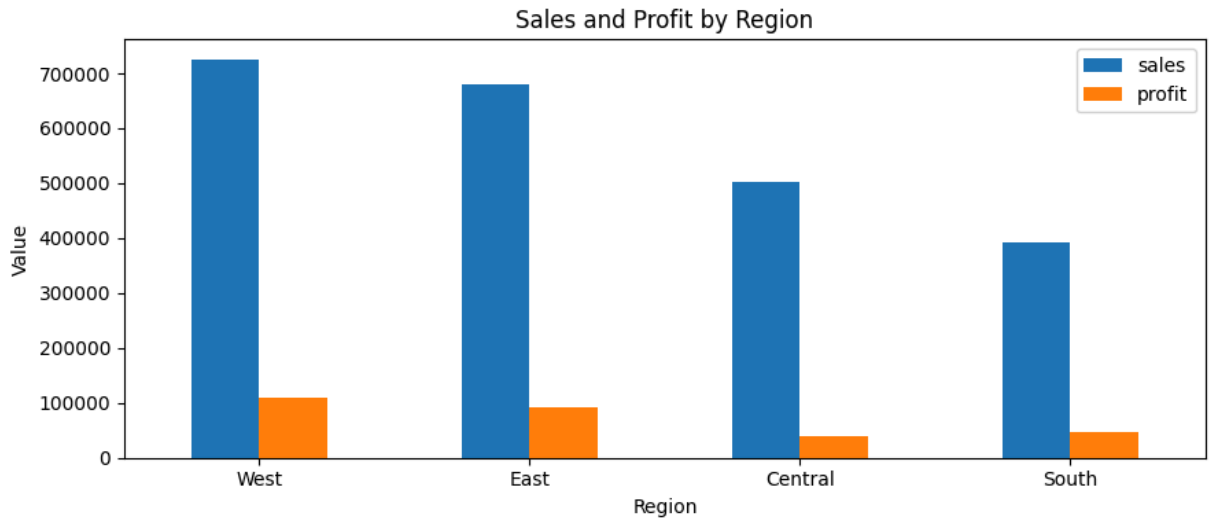
	Region	sales	profit	orders	avg_discount
3	West	725,457.82	108,418.45	1611	0.11
1	East	678,781.24	91,522.78	1401	0.15
0	Central	501,239.89	39,706.36	1175	0.24
2	South	391,721.91	46,749.43	822	0.15

```
Segment performance
```

	Segment	sales	profit	orders	avg_discount
0	Consumer	1,161,401.34	134,119.21	2586	0.16
1	Corporate	706,146.37	91,979.13	1514	0.16
2	Home Office	429,653.15	60,298.68	909	0.15

In [86]:

```
ax = region_perf.plot(
    x="Region",
    y=["sales", "profit"],
    kind="bar",
    figsize=(9, 4),
    title="Sales and Profit by Region",
    rot=0
)
ax.set_ylabel("Value")
plt.tight_layout()
plt.show()
```



13. Discount and profitability association

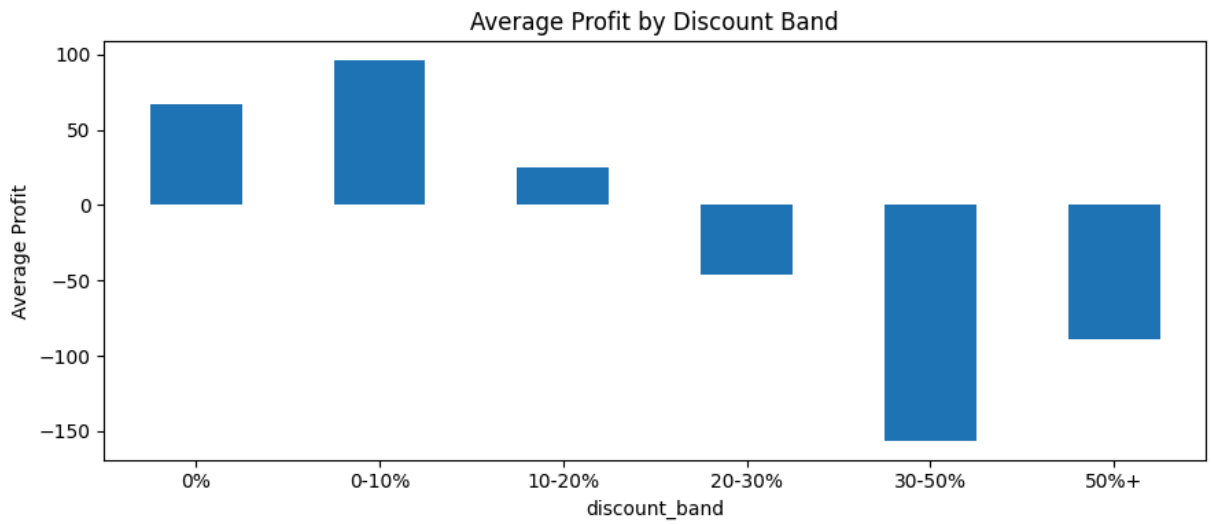
This section is descriptive only.
Any pattern here is an association, not proof that discounting causes profitability outcomes.

In [87]:
```
discount_band = pd.cut(
    df_clean["Discount"],
    bins=[-0.001, 0, 0.1, 0.2, 0.3, 0.5, 1.0],
    labels=["0%", "0-10%", "10-20%", "20-30%", "30-50%", "50%+"]
)

discount_perf = (
    df_clean.assign(discount_band=discount_band)
    .groupby("discount_band", dropna=False, as_index=False)
    .agg(
        rows=("Row ID", "count"),
        sales=("Sales", "sum"),
        profit=("Profit", "sum"),
        avg_profit=("Profit", "mean")
    )
)

discount_perf
```

Out[87]:

	discount_band	rows	sales	profit	avg_profit
0	0%	4798	1,087,908.47	320,987.60	66.90
1	0-10%	94	54,369.35	9,029.18	96.06
2	10-20%	3709	792,152.89	91,756.30	24.74
3	20-30%	227	103,226.65	-10,369.28	-45.68
4	30-50%	310	195,314.76	-48,447.73	-156.28
5	50%+	856	64,228.74	-76,559.05	-89.44

In [88]:
```
ax = discount_perf.plot(
    x="discount_band",
    y="avg_profit",
    kind="bar",
    legend=False,
    figsize=(9, 4),
    title="Average Profit by Discount Band",
    rot=0
)
ax.set_ylabel("Average Profit")
plt.tight_layout()
plt.show()
```



In [89]:
```
corr = df_clean[["Sales", "Quantity", "Discount", "Profit", "ship_delay_days"]].cor
corr
```

Out[89]:

	Sales	Quantity	Discount	Profit	ship_delay_days
Sales	1.00	0.20	-0.03	0.48	-0.01
Quantity	0.20	1.00	0.01	0.07	0.02
Discount	-0.03	0.01	1.00	-0.22	0.00
Profit	0.48	0.07	-0.22	1.00	-0.00
ship_delay_days	-0.01	0.02	0.00	-0.00	1.00

14. Discount vs Profit — Extended Analysis

In [90]:
```
revenue_share_checks = {
    "category_top_share_pct": (
        category_perf["sales"].max() / category_perf["sales"].sum() * 100
    ),
    "top_2_regions_share_pct": (
        region_perf.sort_values("sales", ascending=False).head(2)["sales"].sum()
        / region_perf["sales"].sum() * 100
```

```
    ),
    "top_2_segments_share_pct": (
        segment_perf.sort_values("sales", ascending=False).head(2)["sales"].sum()
        / segment_perf["sales"].sum() * 100
    ),
    "top_20pct_subcategories_share_pct": top_20pct_subcats["sales_share_pct"].sum()
}

revenue_share_checks
```

Out[90]:

```
{'category_top_share_pct': np.float64(36.39882116754926),
 'top_2_regions_share_pct': np.float64(61.12826652505324),
 'top_2_segments_share_pct': np.float64(81.29666604582893),
 'top_20pct_subcategories_share_pct': np.float64(47.417068129503875)}
```

Interpretation boundary:

This is a concentration-based descriptive proxy, not a formal statistical variance decomposition.

15. Time-based descriptive view

In [91]:

```
monthly_perf = (
    df_clean.groupby("month", as_index=False)
    .agg(
        sales=("Sales", "sum"),
        profit=("Profit", "sum"),
        orders=("Order ID", "nunique")
    )
    .sort_values("month")
)

monthly_perf.head()
```

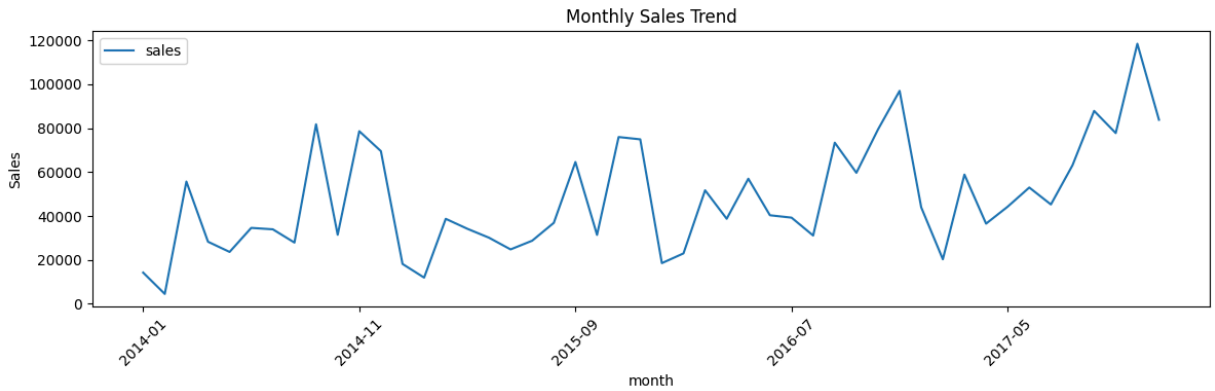
Out[91]:

	month	sales	profit	orders
0	2014-01	14,236.90	2,450.19	32
1	2014-02	4,519.89	862.31	28
2	2014-03	55,691.01	498.73	71
3	2014-04	28,295.35	3,488.84	66
4	2014-05	23,648.29	2,738.71	69

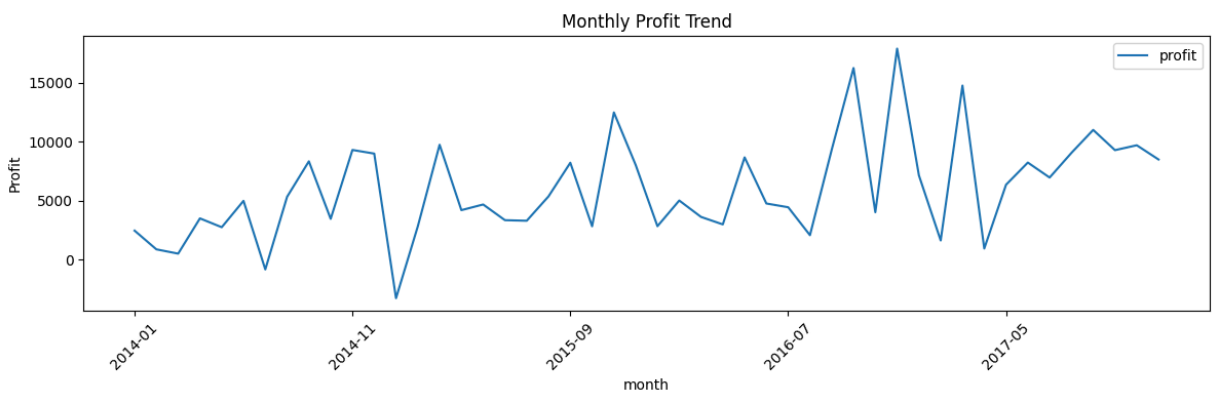
In [92]:

```
ax = monthly_perf.plot(
    x="month",
    y="sales",
    kind="line",
    figsize=(12, 4),
    title="Monthly Sales Trend"
)
```

```
ax.set_ylabel("Sales")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



In [93]:
```
ax = monthly_perf.plot(
    x="month",
    y="profit",
    kind="line",
    figsize=(12, 4),
    title="Monthly Profit Trend"
)
ax.set_ylabel("Profit")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



16. Evidence tables for portfolio reporting

In [94]:
```
evidence_top = {
    "top_category_by_sales": category_perf.sort_values("sales", ascending=False).il
    "top_category_sales": float(category_perf.sort_values("sales", ascending=False)
    "top_region_by_sales": region_perf.sort_values("sales", ascending=False).iloc[0
    "lowest_subcategory_by_profit": subcat_perf.sort_values("profit", ascending=Tru
    "lowest_subcategory_profit": float(subcat_perf.sort_values("profit", ascending=
    "negative_profit_rows": int((df_clean["Profit"] < 0).sum())
}
evidence_top
```

Out[94]:
```
{'top_category_by_sales': 'Technology',
 'top_category_sales': 836154.033,
 'top_region_by_sales': 'West',
 'lowest_subcategory_by_profit': 'Tables',
 'lowest_subcategory_profit': -17725.4811,
 'negative_profit_rows': 1871}
```

17. What the data supports

Use this section to write evidence-based statements only, for example:

- which category leads in sales
- which region contributes the most profit
- which sub-categories show the largest losses
- whether higher discount bands are associated with lower average profit

Do not write causal statements here.

18. What the data does not support

This notebook does **not** support:

- claims that discounts cause losses
- claims that shipping delay causes low profit
- forecasts of future sales
- operational prescriptions without ACT-stage evidence mapping

19. Validation checks completed

- Dataset load completed
- Schema inspected
- Missingness profiled
- Duplicate checks run
- Range checks run
- Date parsing validated
- Explicit processing steps documented
- Summary metrics reproduced from processed data

20. Final notebook closeout

Recommended follow-up artifacts

- `01_PREPARE/2026-03-25__superstore__prepare.md`

- `02_PROCESS/2026-03-25__superstore__process_log.md` if you want full transformation logging
- `03_ANALYZE/2026-03-25__superstore__analysis.md`

Fully addressed

- Revenue and profit leaders by category, region, and segment
- Loss-making concentration at sub-category level
- Reproducible aggregation tables

Addressed after patch

- Top 20% revenue contributors
- Negative-profit patterns within finer segment layers
- Revenue concentration proxy for the 60% success criterion

Not claimed

- Causal drivers
- Forecasting
- Action recommendations

Remaining limitations

- Synthetic/static dataset context
- No external benchmarks
- No inferential design for causality
- No formal sensitivity analysis beyond descriptive checks